



Computer network

Introduction

Ethernet LAN



Content

- LAN def – Ethernet cable - SW
- PC – SW - PC
- Routing & router (LANs connection)



LAN

- Connect devices in a small area
- Segment length

Table 2-1 *Ethernet Segment Distance Limitations*

Ethernet Specification	Description	Segment Length
10BASE-T	10-Mbps Ethernet over twisted-pair	100 m
10BASE-FL	10-Mbps over fiber-optic cable	2000 m
100BASE-TX	100-Mbps Ethernet over twisted-pair	100 m
100BASE-FX	Fast Ethernet, still 100-Mbps, over fiber-optic cable	400 m
1000BASE-T	Gigabit Ethernet, 1000-Mbps, over twisted-pair	100 m
1000BASE-LX	Gigabit Ethernet over fiber-optic cable	550 m if 62.5-micron (μ) or 50- μ multimode fiber; 10 km if 10- μ single-mode fiber
1000BASE-SX	Gigabit Ethernet over fiber-optic cable	250 m if 62.5- μ multimode fiber; 550 m if 50- μ multimode fiber
1000BASE-CX	Gigabit Ethernet over copper cabling	25 m

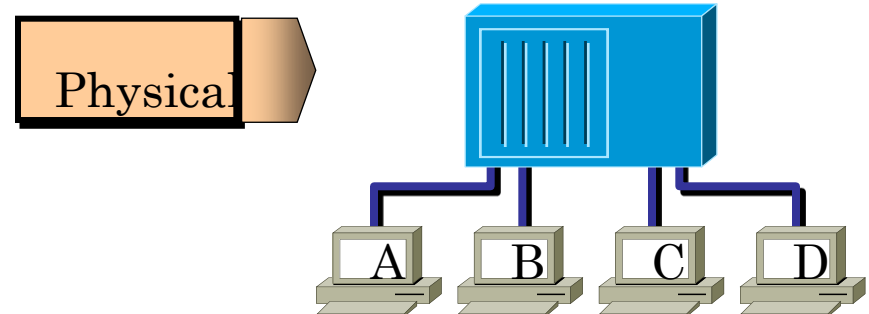
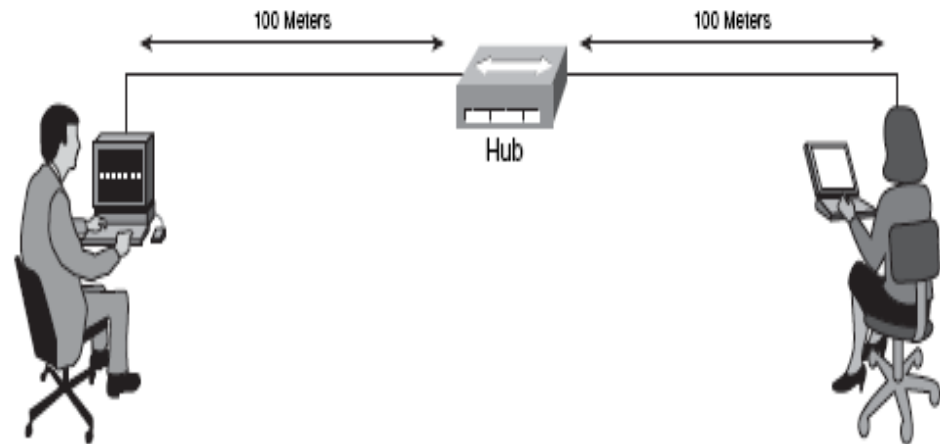


Extending LAN with Hub



- Hub/ repeater can overcome the distance limitation in LAN
- Hub can have multiple ports to connect to a number of devices

Figure 2-1 Extending the Segment Link with a Hub





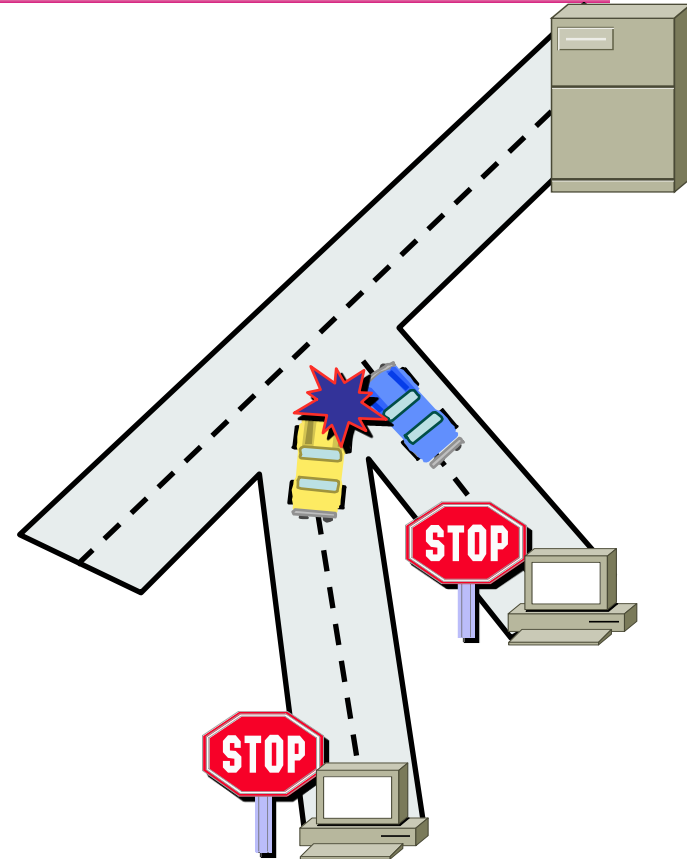
Hub: one collision domain



- More end stations means more collisions
- CSMA/CD is used

(Carrier Sense Multiple Access with Collision Detect)

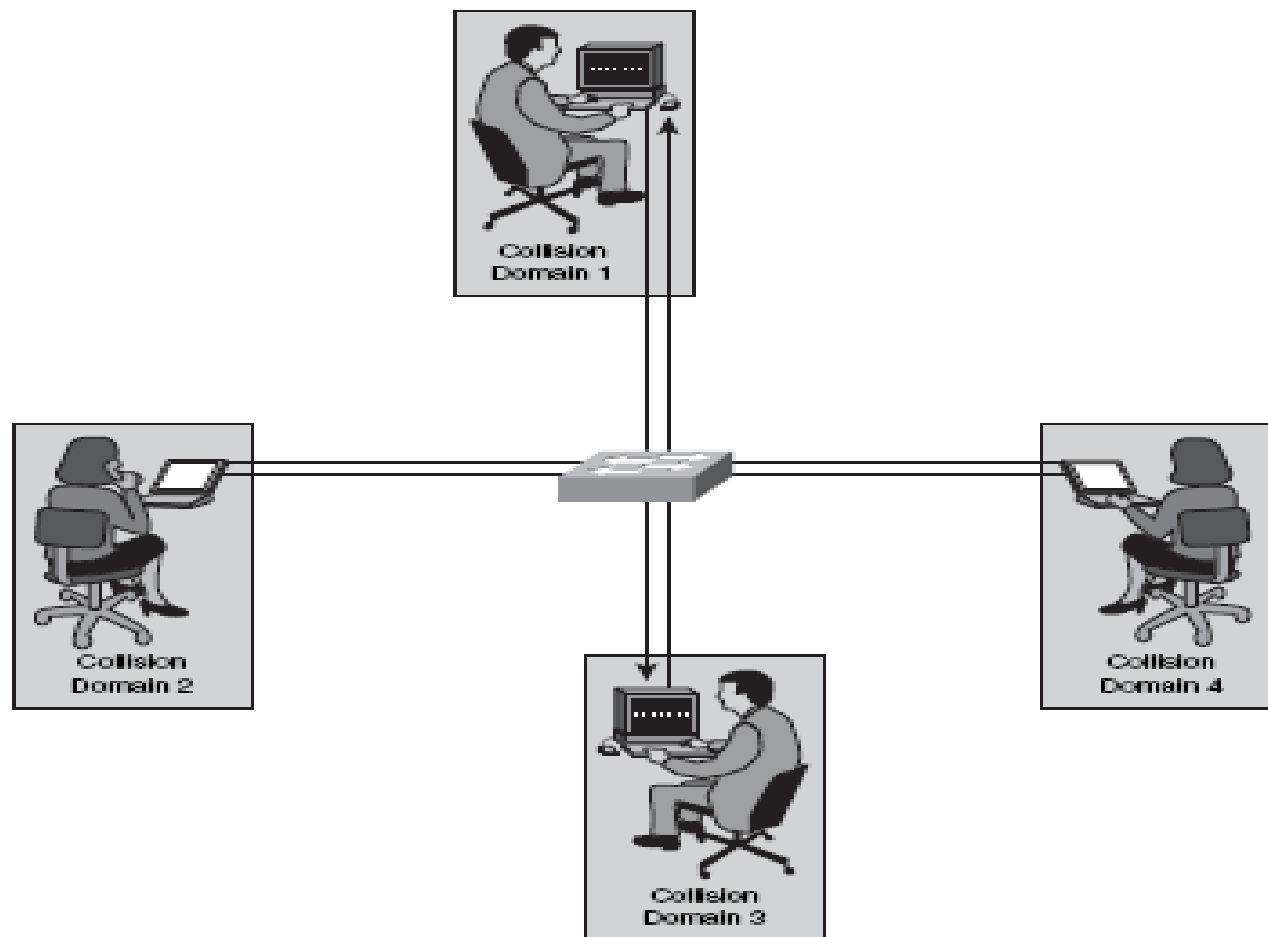
- Collision domain is the network segment that share the same bandwidth





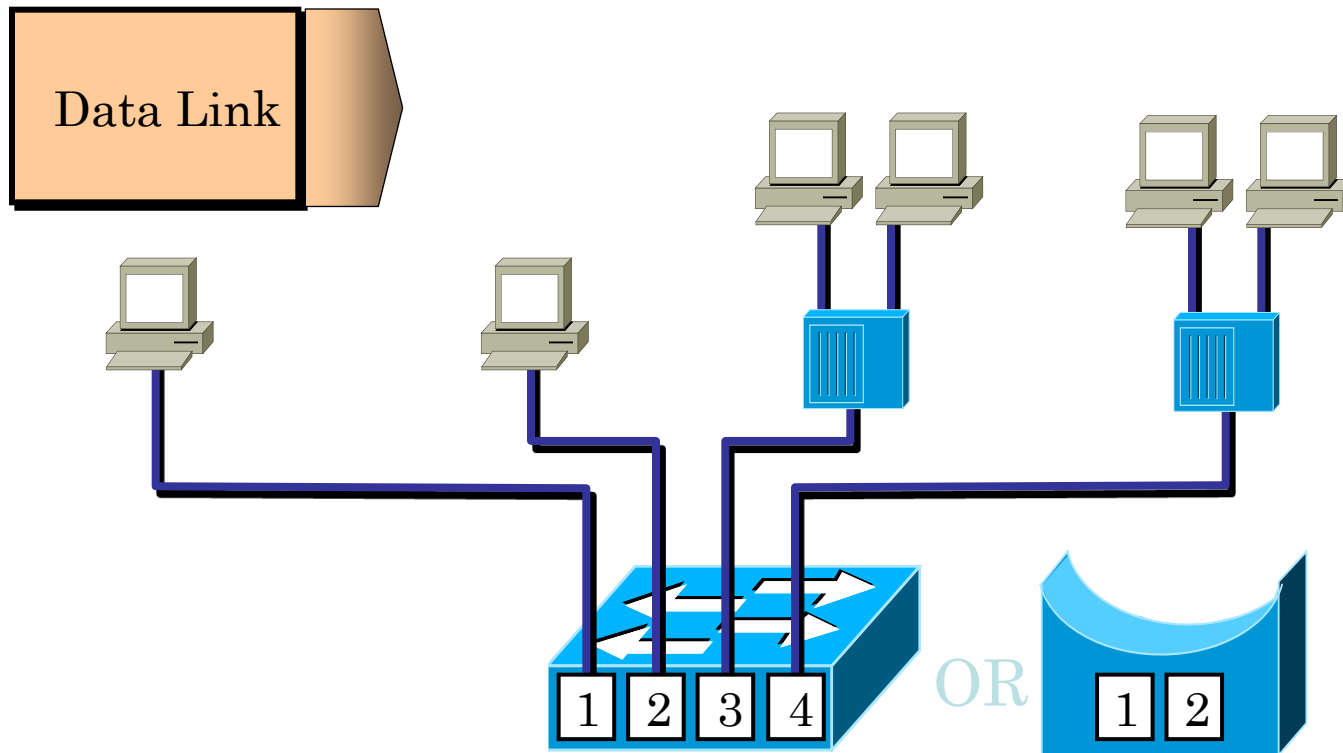
Create multiple Collision domain with Switch

Figure 2-3 *Creating Multiple Collision Domains Using a Switch*





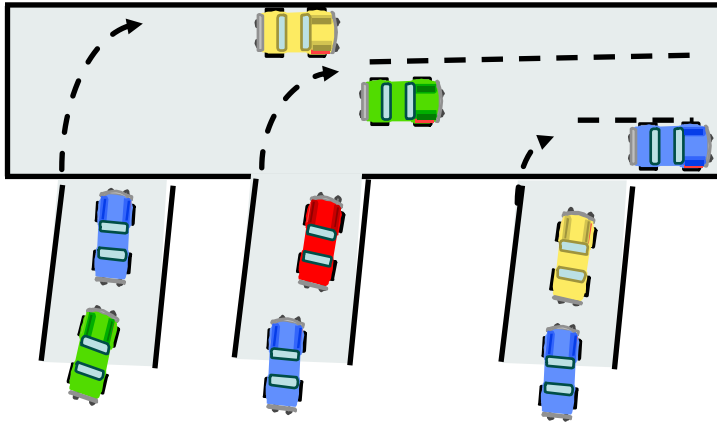
Switches and bridge operate at data link layer



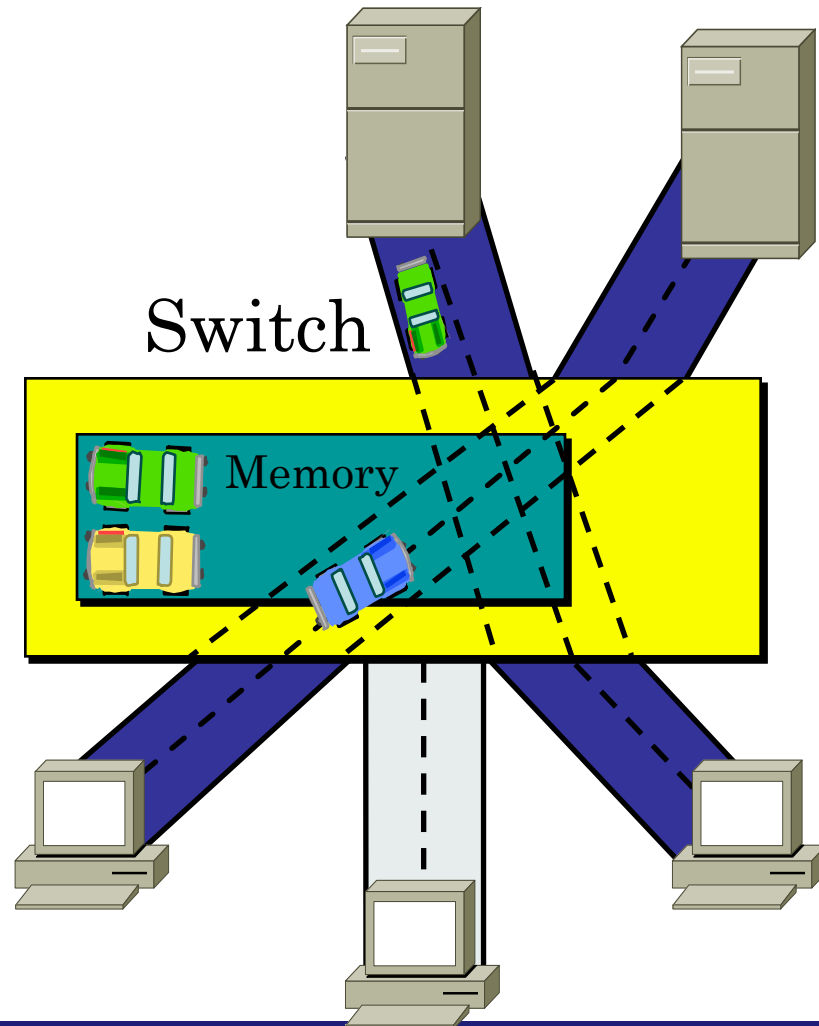
- Each segment has its own collision domain
- All segments are in the same broadcast domain



switches



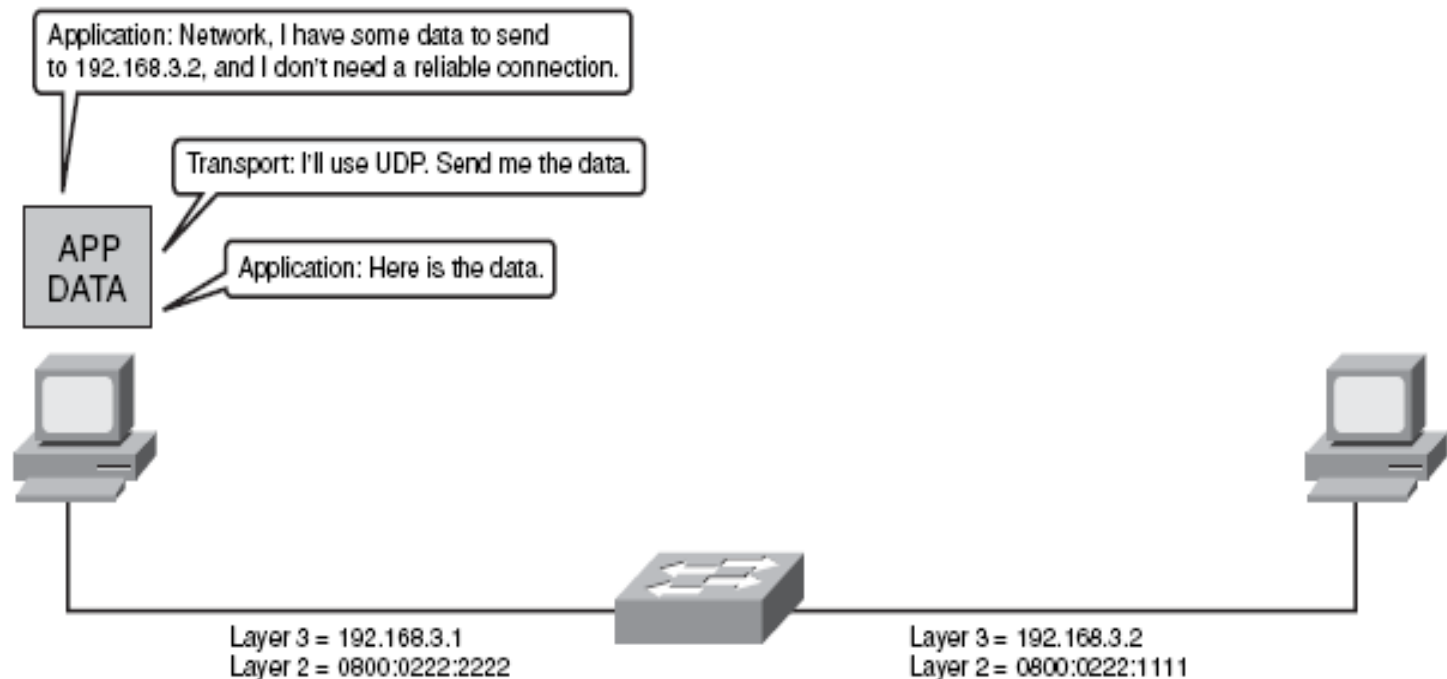
- Each segment has its own collision domain
- Broadcasts are forwarded to all segments





Packet delivery through SW

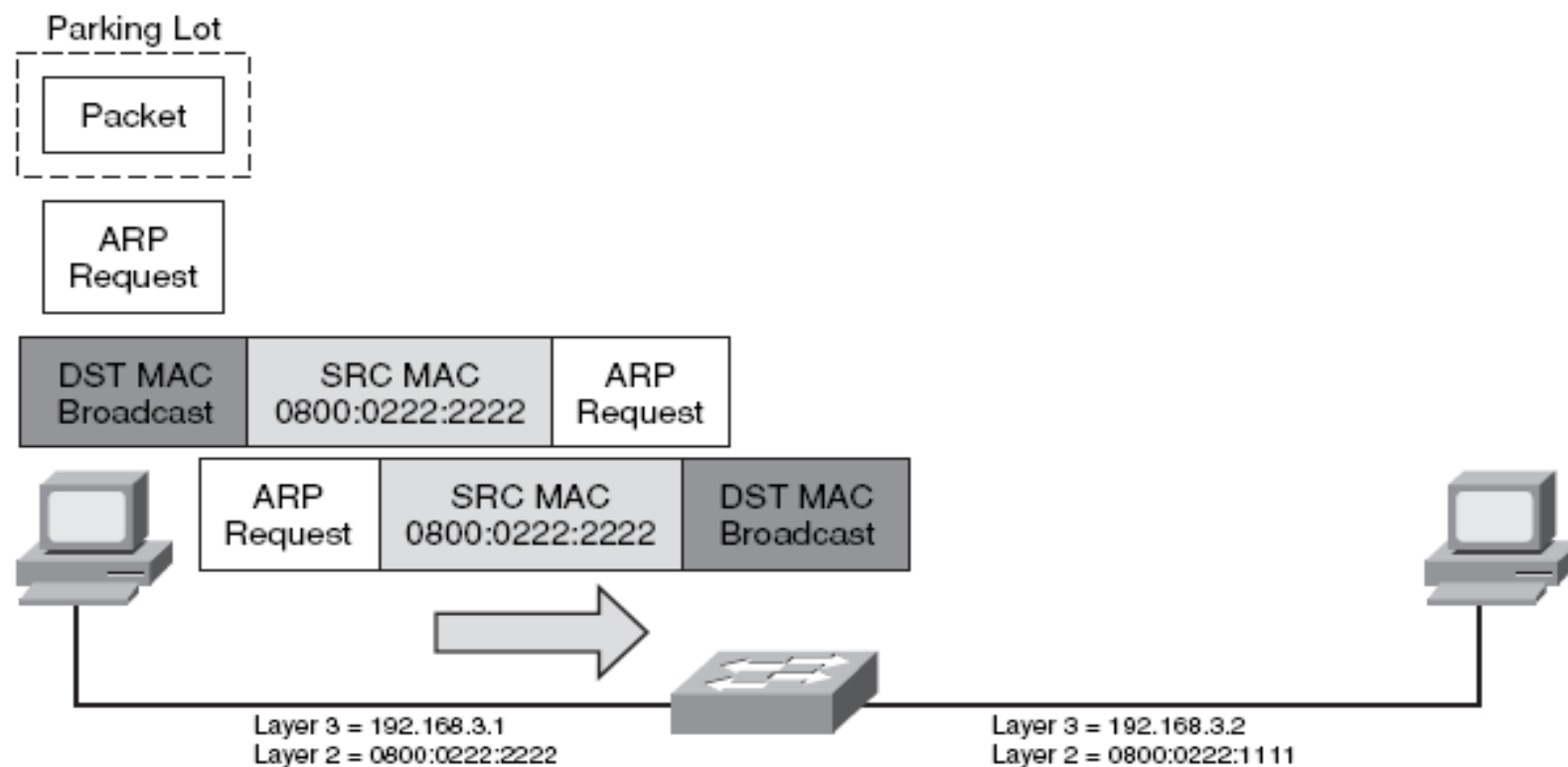
Figure 2-4 *Host Sending Data*





ARP request

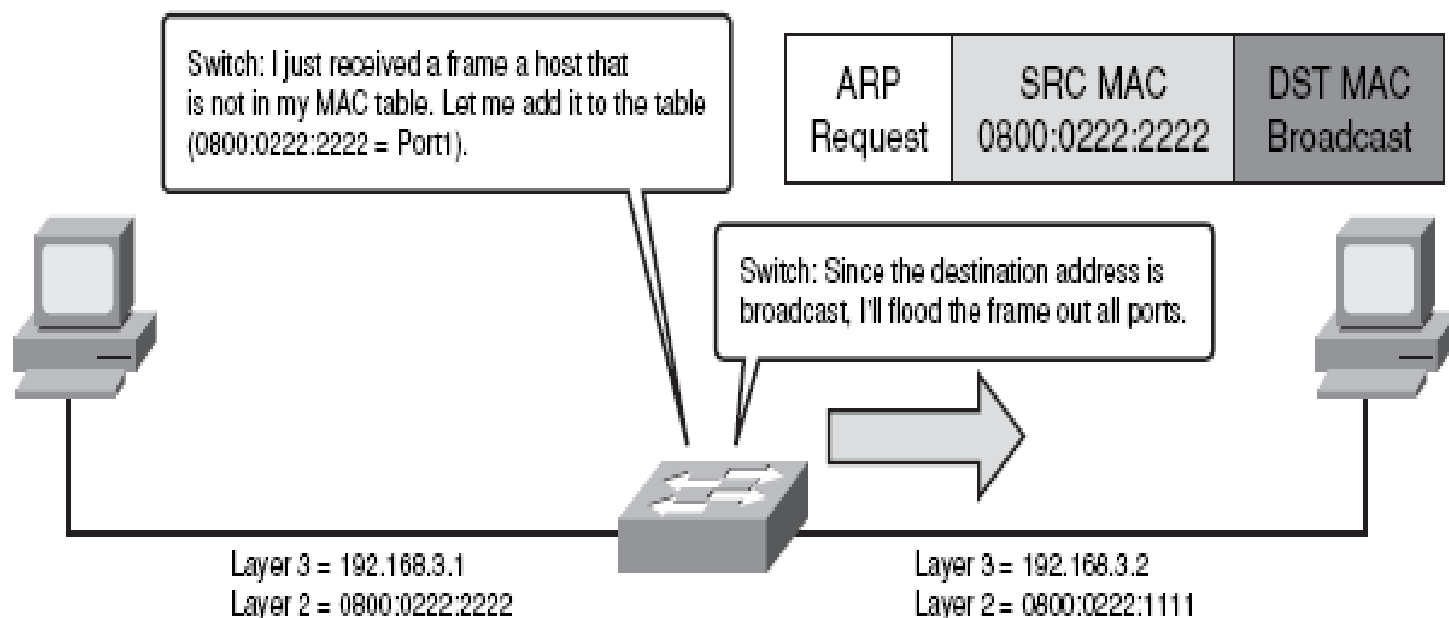
Figure 2-7 *Sending the ARP Request*





SW Learning & Forwarding

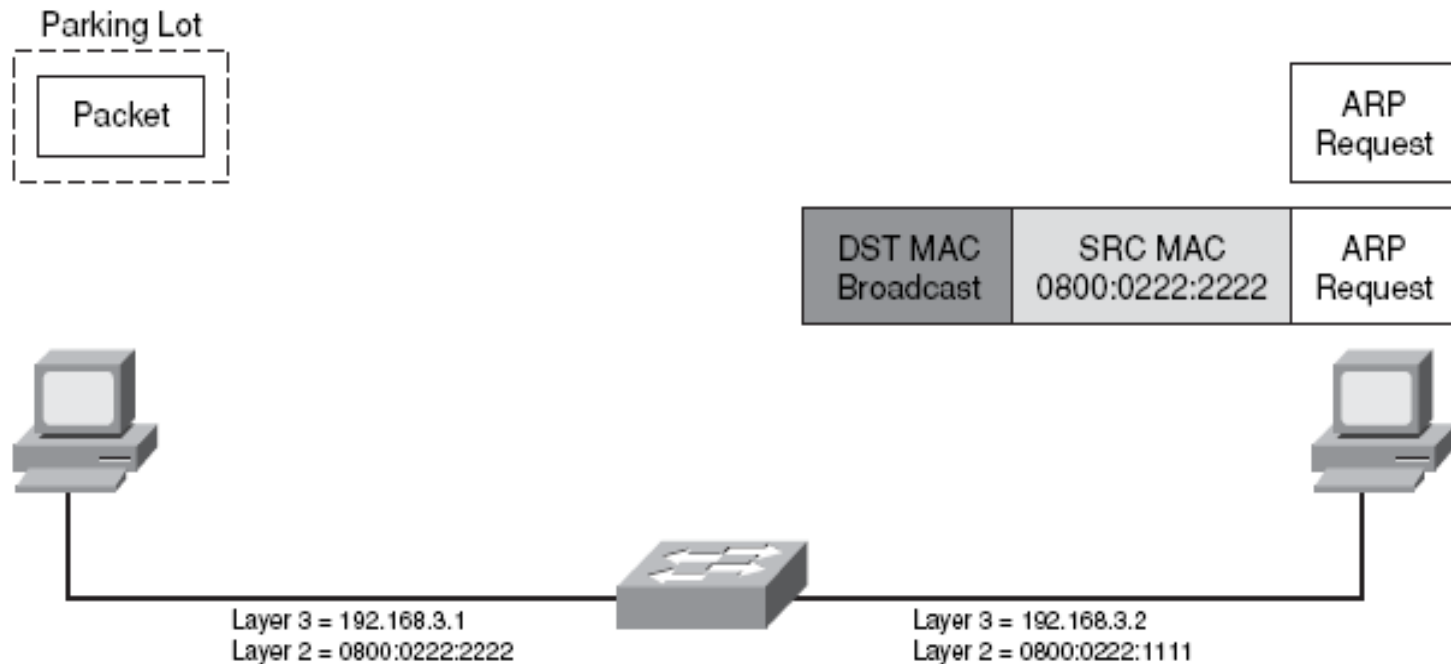
Figure 2-8 *Switch Learning and Forwarding*





Host receive ARP request

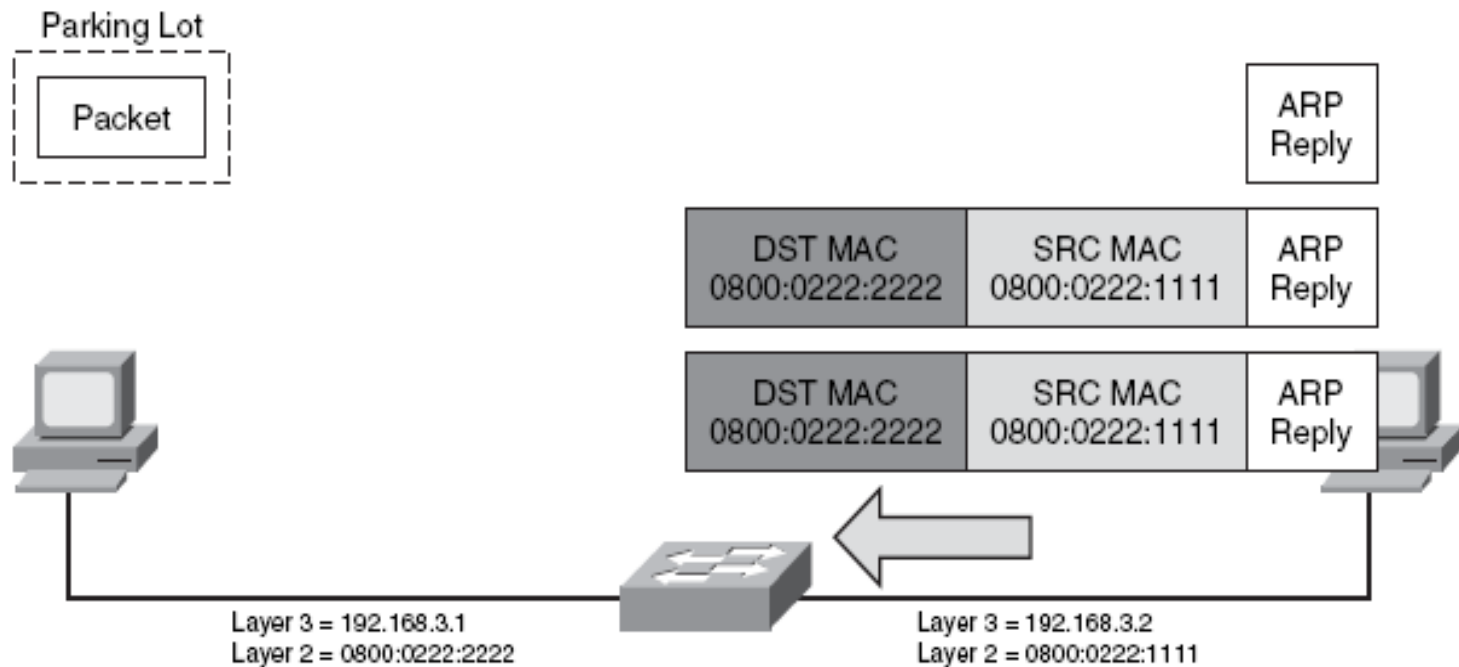
Figure 2-9 *Host Receives ARP Request*





ARP reply

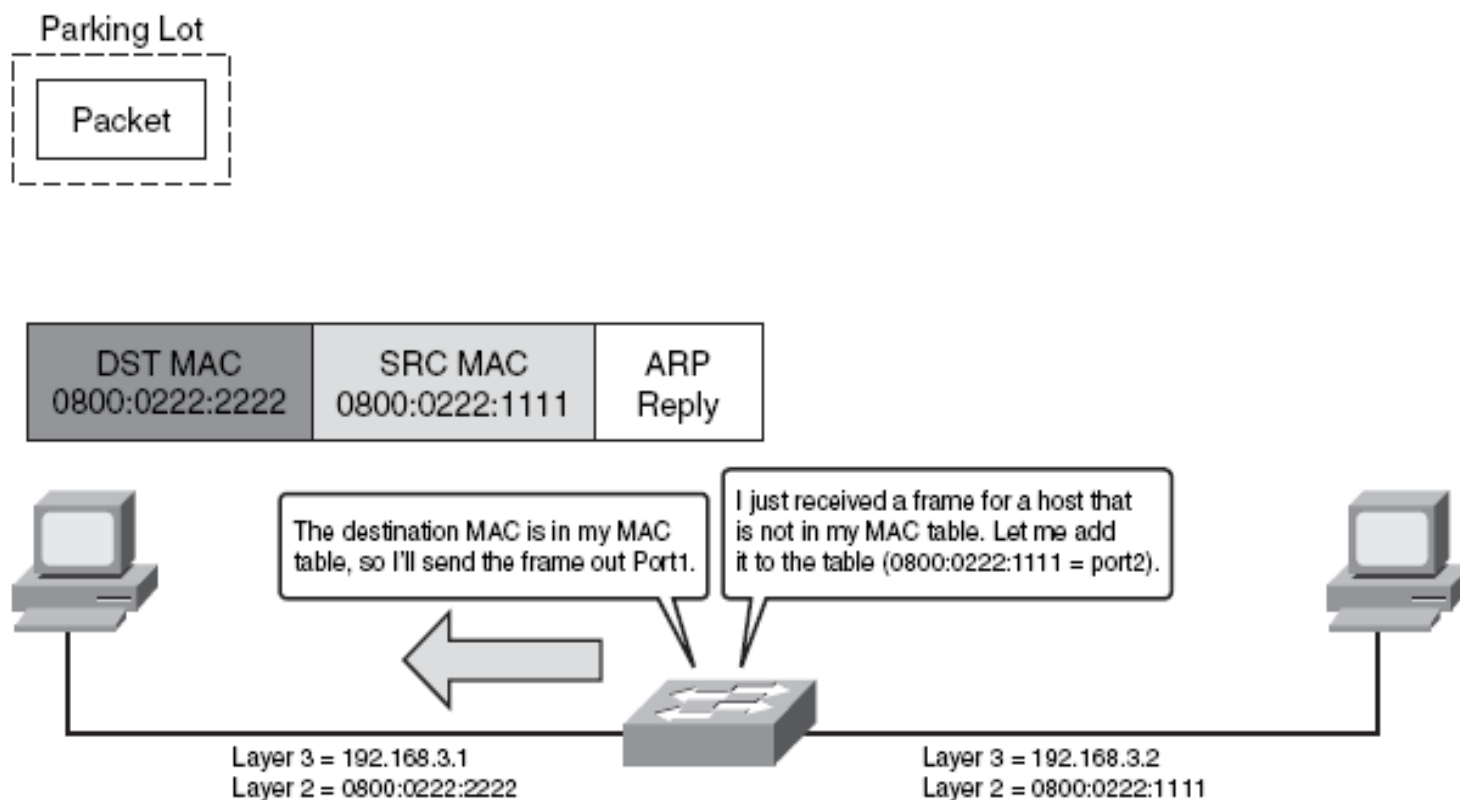
Figure 2-10 *Host Responds to ARP Request*





ARP reply received

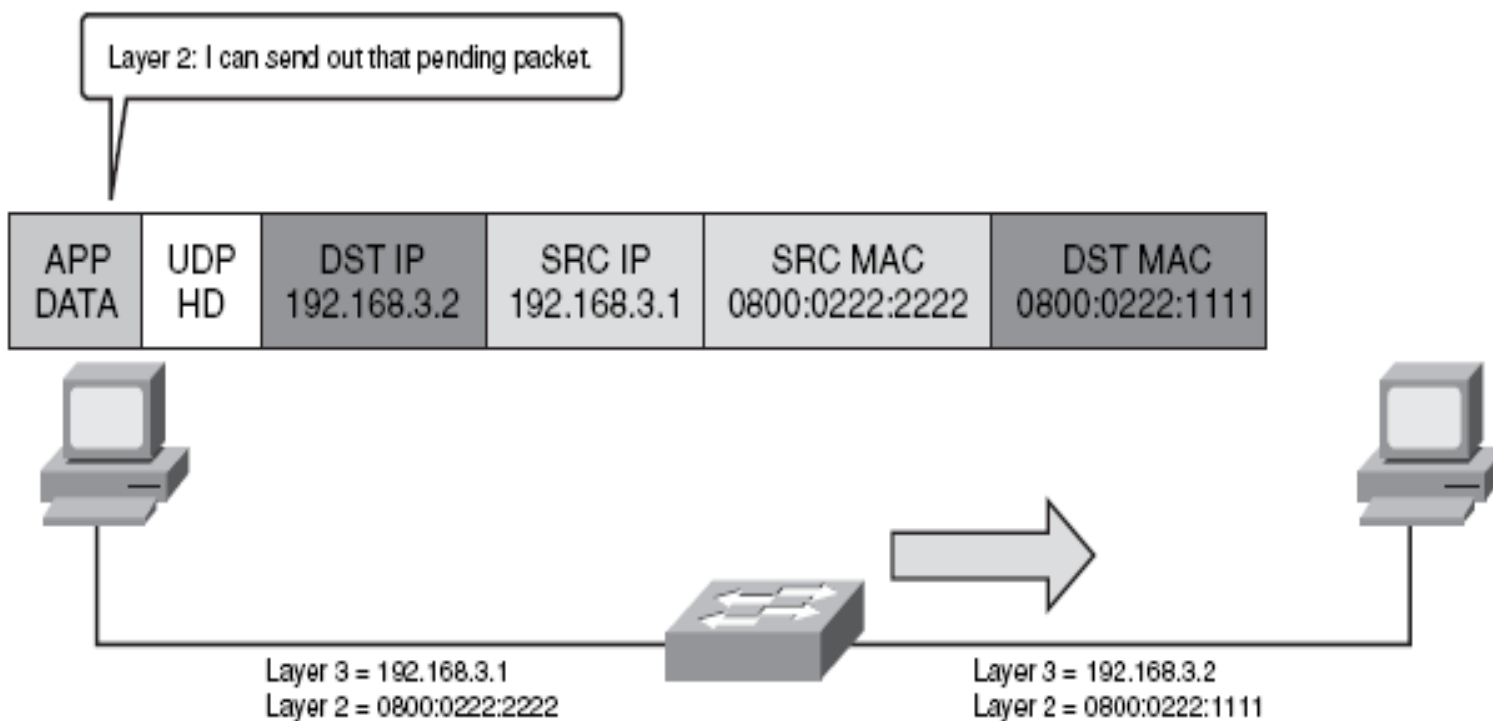
Figure 2-11 *Host Receives ARP Response*





SW forwarding frame

Figure 2-13 *Switch Forwards Frame*

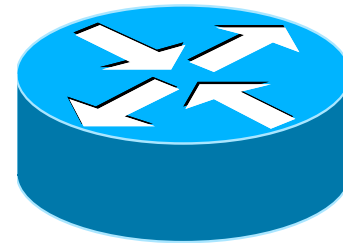




Router

- Router is a special computer device
- Components
 - CPU
 - Motherboard
 - RAM
 - ROM
- Routers determine the path for transmitting data from one network to another
- Ports

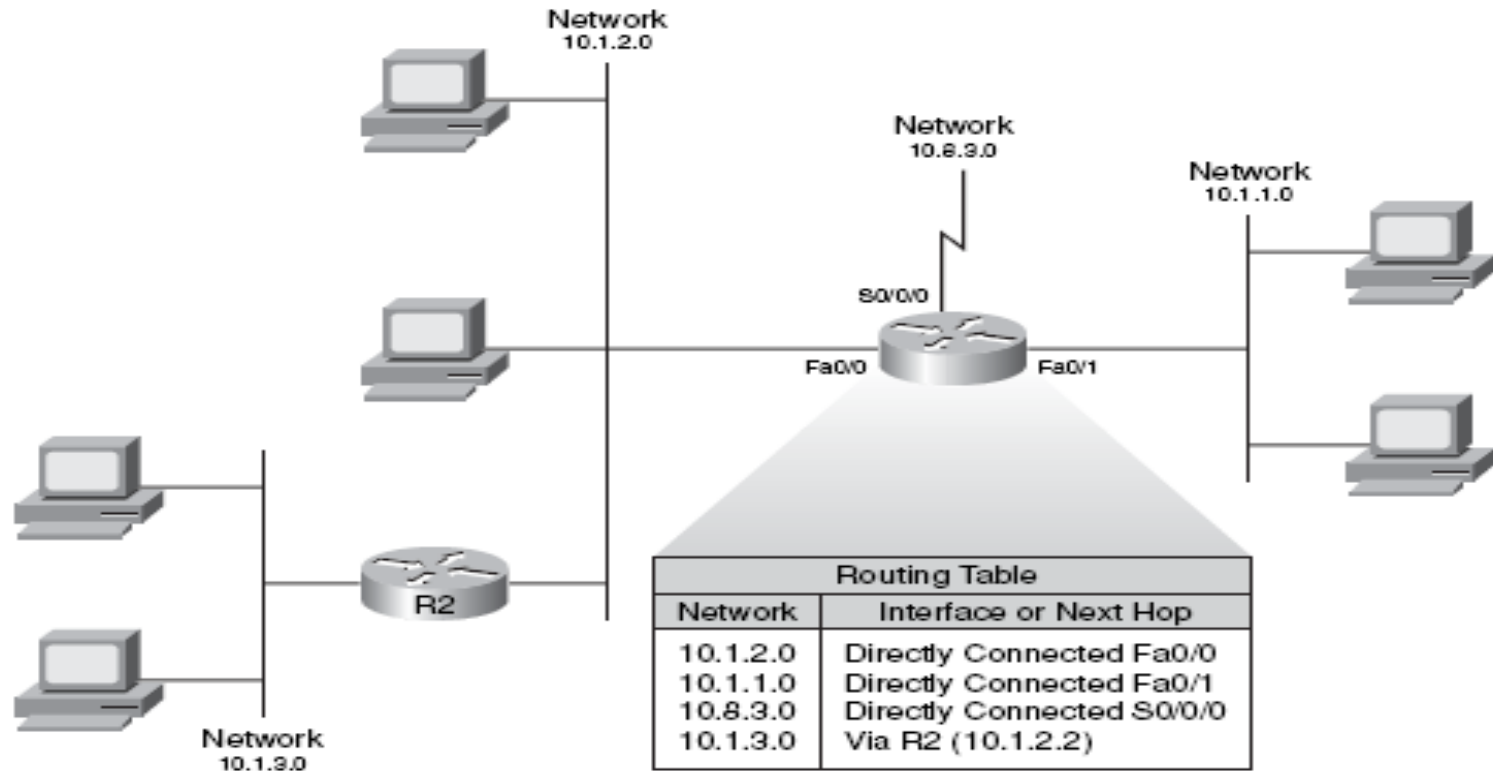
Figure 4-1 Cisco 2800 Series Routers





Routing table

- Router builds routing table to identify known networks and know how to reach them





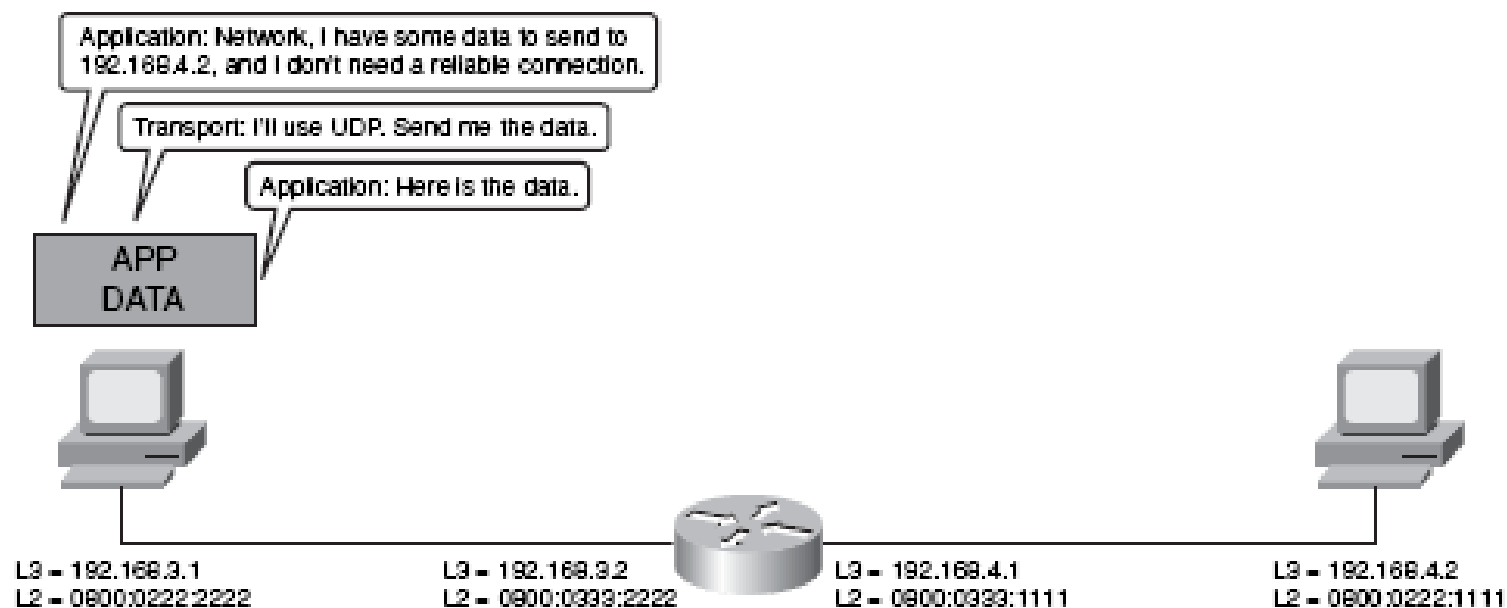
Deliver an IP packet

- Similar to send a letter through mail delivery service
- Key is to know source & destination address
- Host will sent non-local packet to default gateway



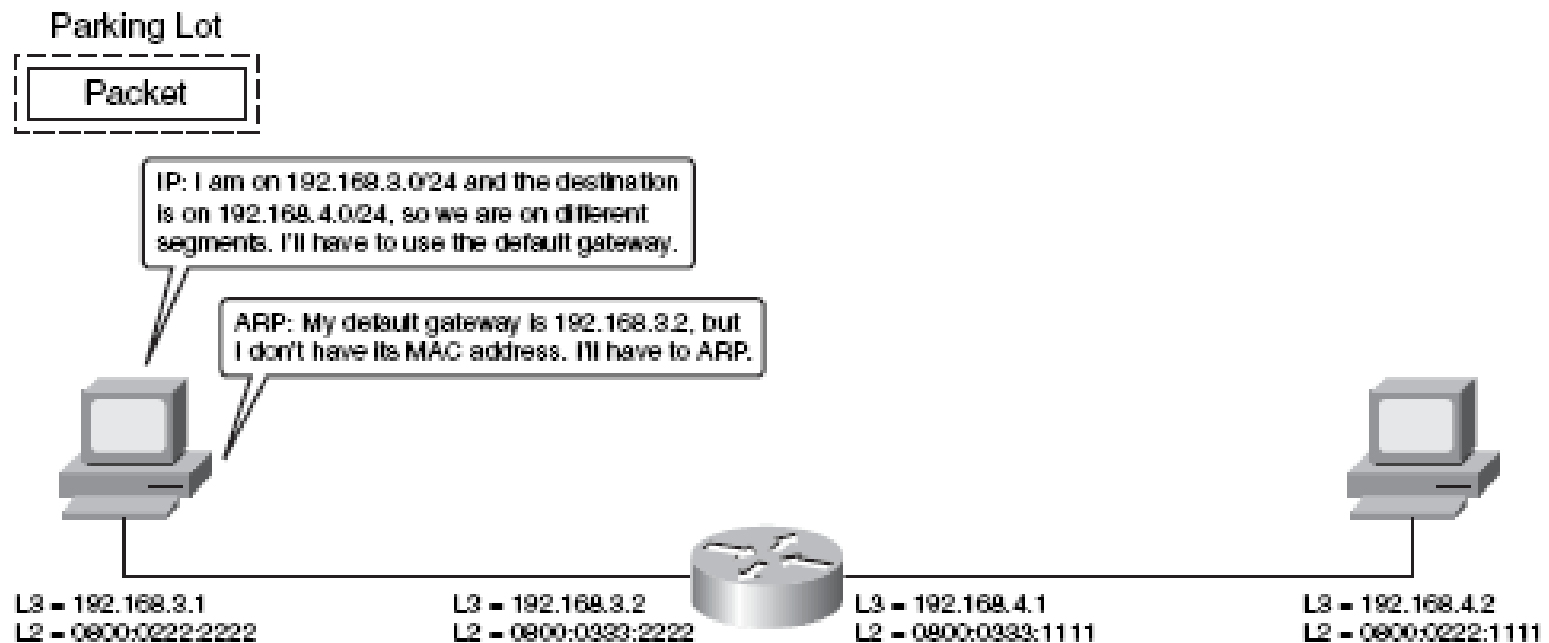
Packet delivery

Figure 4-31 *Application Sends Data to Transport Stack*



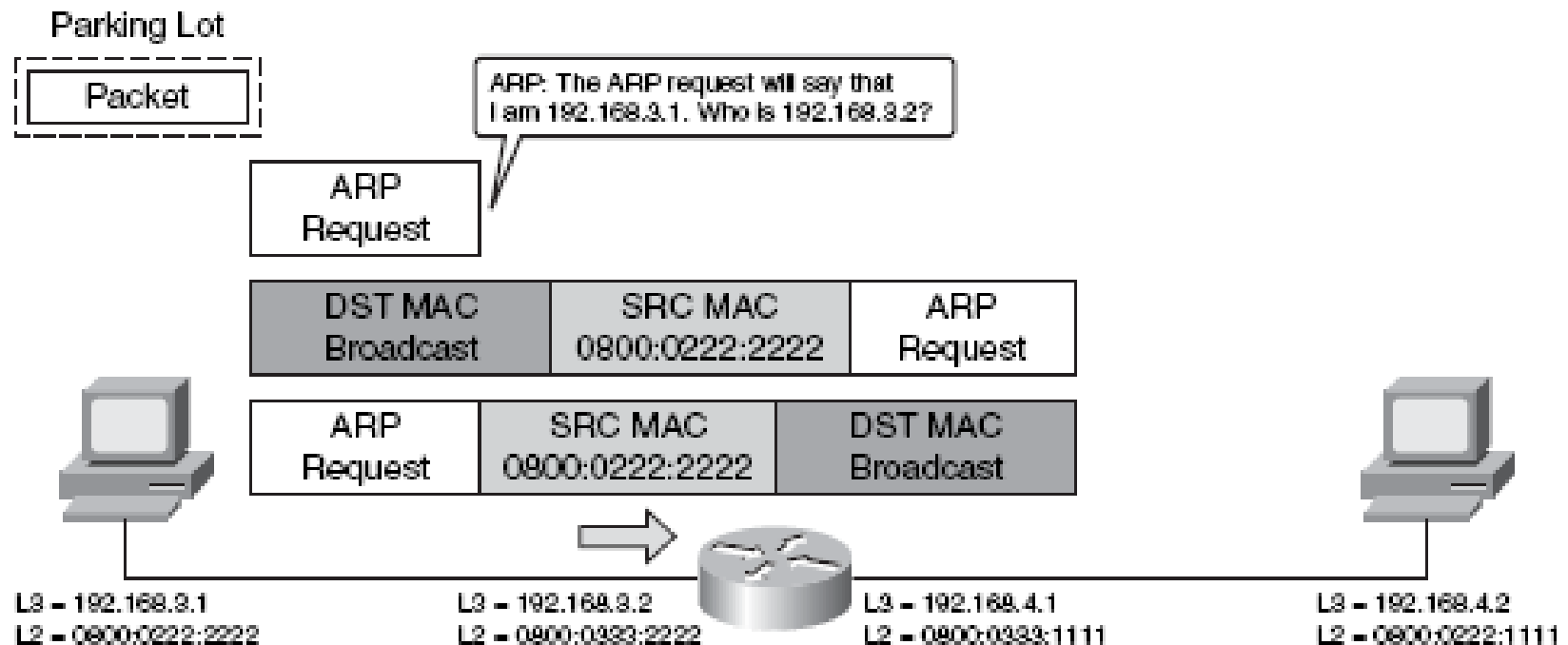


Default gateway



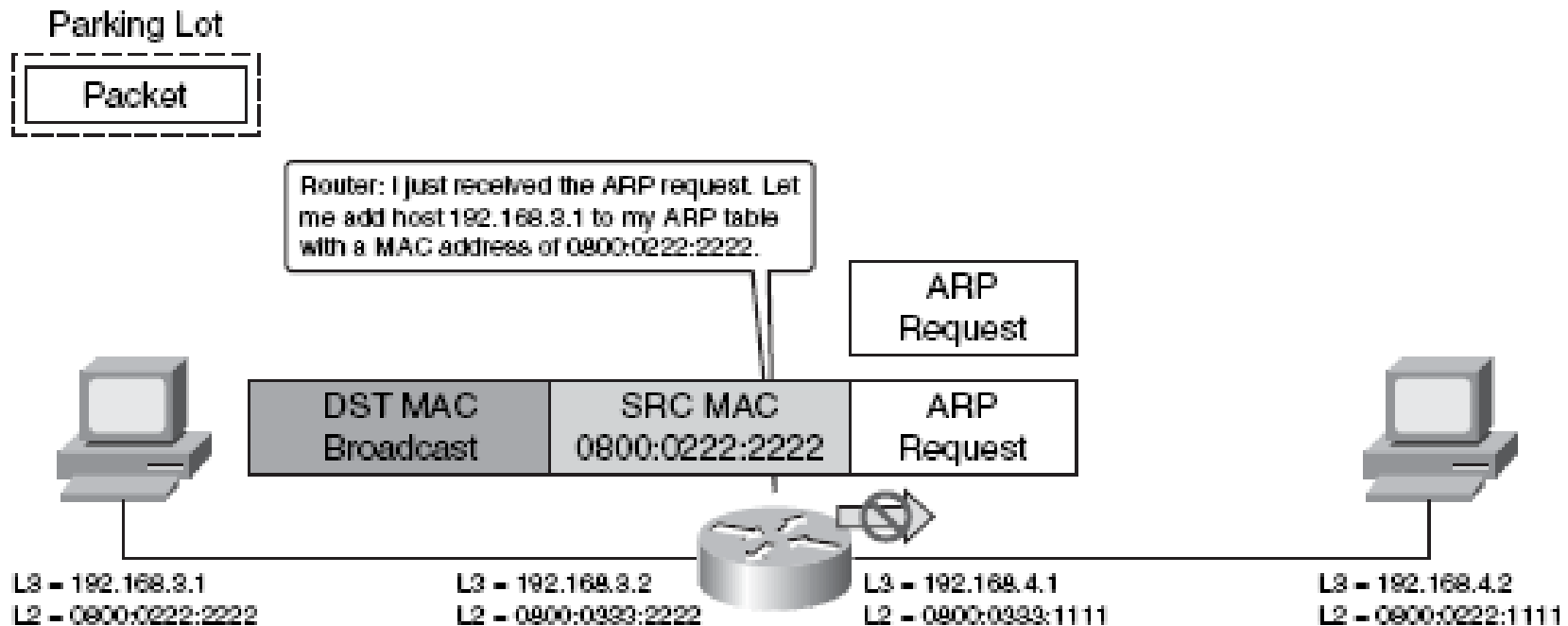


ARP request to router



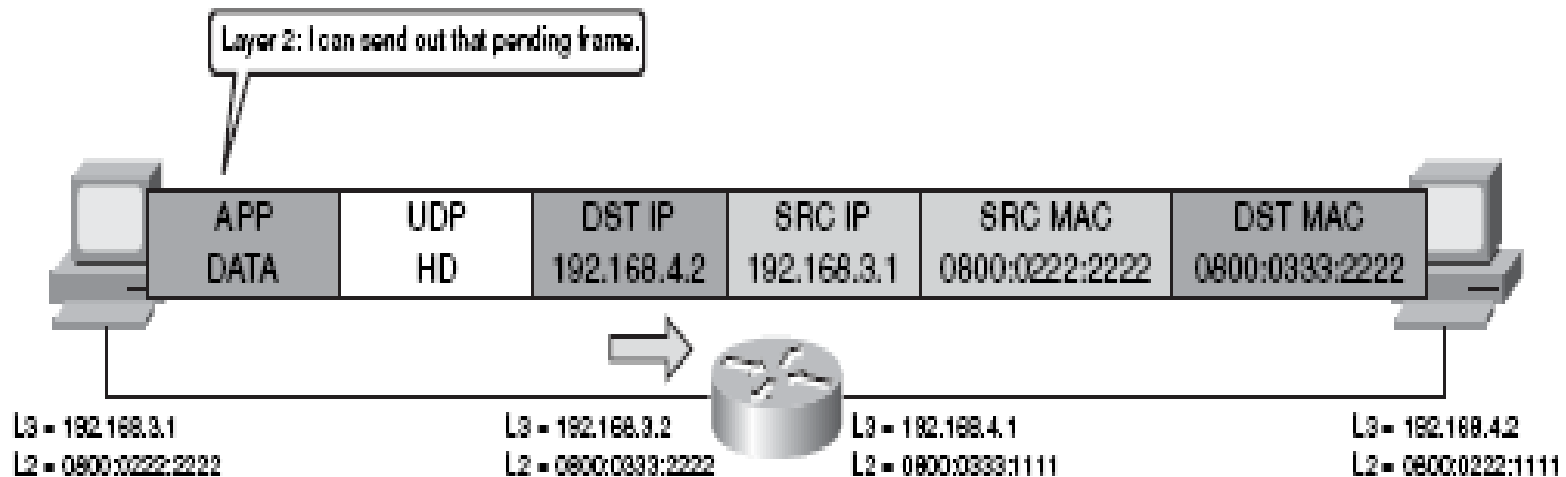


Router receives ARP request



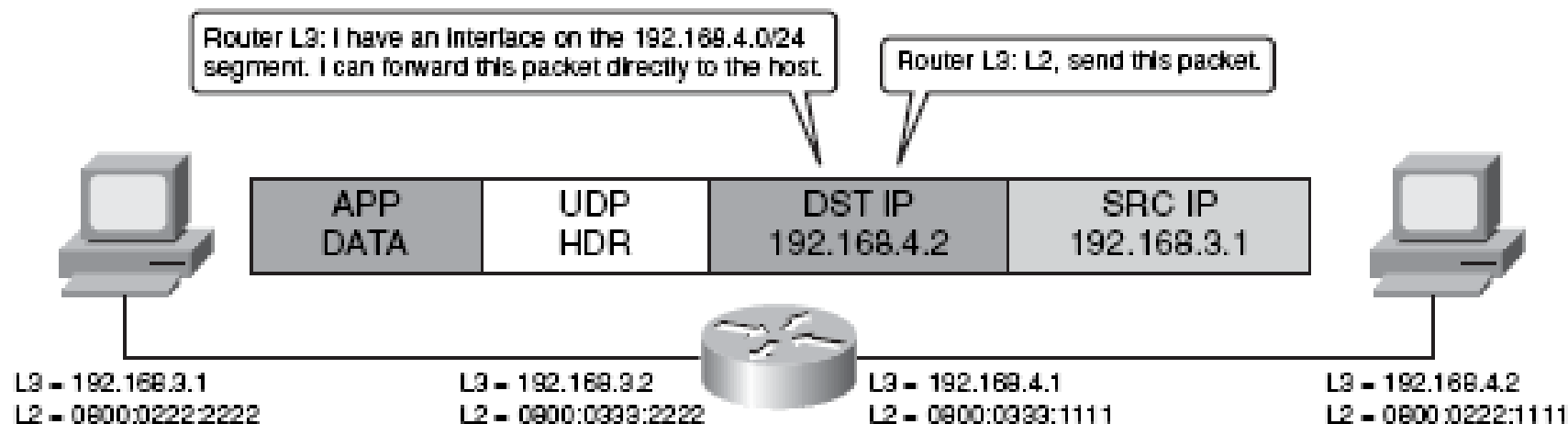


Send data frame to router





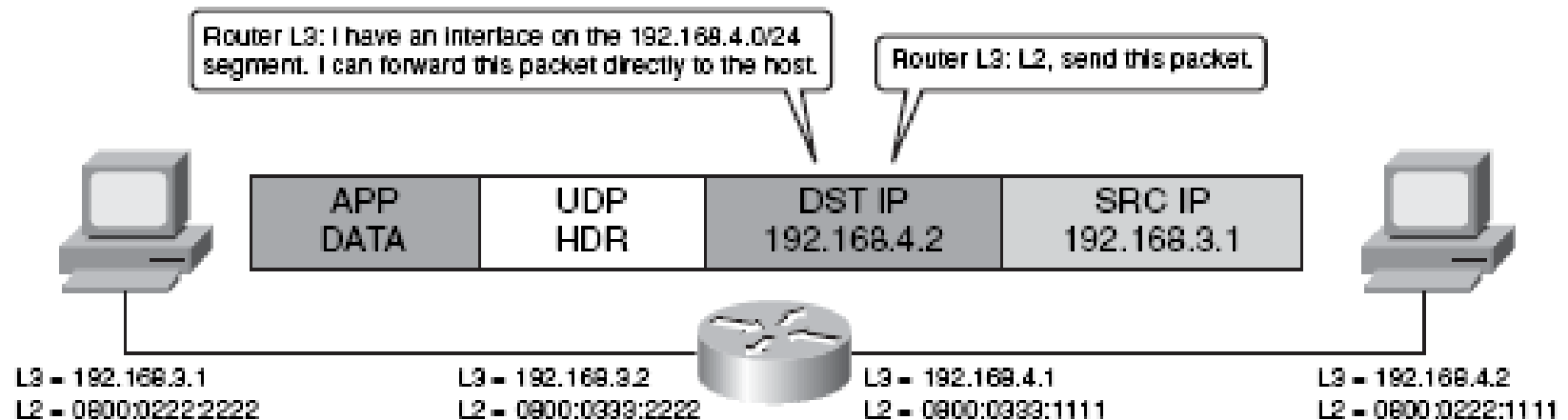
Routing table lookup



Destination	Next Hop	Interface
192.168.3.0/24	Connected	Fa0/0
192.168.4.0/24	Connected	Fa0/1



Router send ARP request



Destination	Next Hop	Interface
192.168.3.0/24	Connected	Fa0/0
192.168.4.0/24	Connected	Fa0/1



Router send frame to the destination host

